



RECOVERY & REPAIR

TB-500

TB-500 – 10 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Thymosin Beta-4 (full-length, 43 residues)
SYNONYMS / ALIASES	TB-500; Tbeta4; synthetic thymosin beta-4
CAS NUMBER	77591-33-4
MOLECULAR FORMULA	C212H350N56O78S
MOLECULAR WEIGHT	Approximately 4961 g/mol
AMINO ACID SEQUENCE	SDKPDMAEIEKFDKSKLKKTTETQEKNPLPSKETIEQEQQAGES (43 residues; full-length native human Tbeta4)
SOURCE	Synthetic (solid-phase peptide synthesis); sequence corresponds to native human thymosin beta-4

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	10 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 14-28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Methionine residue is susceptible to oxidation; protect from prolonged air exposure. Avoid repeated freeze-thaw cycles.

STRUCTURAL NOTES

TB-500 is the commercial designation for synthetic thymosin beta-4 (Tbeta4), a 43-amino-acid G-actin sequestering peptide ubiquitously expressed in mammalian cells. This product is the full-length 43-residue native sequence (not a fragment). The active actin-binding region of the peptide is localized to residues 17-23 (LKKTETQ).

SELECTED PEER-REVIEWED REFERENCES

1. Goldstein A.L. et al. - *Thymosin beta-4: actin-sequestering protein moonlights to repair injured tissues* - Trends in Molecular Medicine, 2005 - DOI: 10.1016/j.molmed.2005.07.002
2. Crockford D. et al. - *Thymosin beta-4: structure, function, and biological properties supporting current and future clinical applications* - Annals of the New York Academy of Sciences, 2010 - DOI: 10.1111/j.1749-6632.2010.05607.x
3. Sosne G. et al. - *Thymosin beta 4: a novel regenerative molecule* - Expert Opinion on Biological Therapy, 2007 - DOI: 10.1517/14712598.7.12.1809

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.